Full Stack .NET Developer Interview Questions 2025

C# Fundamentals (20)

Q: What is C#?
A: A modern, object-oriented programming language by Microsoft.

Q: Explain CLR.
A: Common Language Runtime manages code execution.

Q: What is JIT?
A: Just-In-Time compiler converts IL to native code.

Q: What is boxing?
A: Converting value type to object type.

Q: What is unboxing?
A: Extracting value type from object.

Q: Explain nullable types.
A: Types that can hold null values.

Q: What is async/await?
A: Keywords for asynchronous programming.

Q: What is a delegate?
A: Type-safe function pointer.

Q: What is an event?
A: Notification mechanism using delegates.

Q: Explain properties.
A: Class members to encapsulate fields.

Q: What is inheritance?
A: Deriving a class from another.

Q: What is polymorphism?
A: Ability to take multiple forms.

Q: What is encapsulation?
A: Hiding internal details of a class.

Q: What is abstraction?
A: Hiding implementation details.

Q: What is interface?
A: Contract with no implementation.

Q: What is abstract class?
A: Class with abstract methods.

Q: Explain sealed class.
A: Class that cannot be inherited.

Q: What is static class?
A: Class with only static members.

Q: What is extension method?
A: Adds methods to existing types.

Q: What is LINQ?
A: Language Integrated Query for data.

ASP.NET Core & MVC (20)

Q: What is ASP.NET Core?
A: Cross-platform framework for web apps.

Q: Explain middleware.
A: Components that handle requests/responses.

Q: What is MVC?
A: Model-View-Controller design pattern.

Q: What is Razor?
A: Syntax for embedding C# in HTML.

Q: Explain ViewBag.
A: Dynamic object for passing data to view.

Q: Explain ViewData.
A: Dictionary for passing data to view.

Q: Explain TempData.
A: Stores data for one request.

Q: What is TagHelper?
A: Server-side code in Razor views.

Q: What is routing?
A: Mapping URLs to actions.

Q: Explain attribute routing.
A: Define routes using attributes.

Q: What is dependency injection?
A: Technique to achieve loose coupling.

Q: What is Startup.cs?
A: Configures services and middleware.

Q: Explain filters.
A: Logic before/after action execution.

Q: What is action result?
A: Return type of controller actions.

Q: Explain partial view.
A: Reusable view component.

Q: What is layout view?
A: Defines common structure for views.

Q: Explain model binding.
A: Maps request data to action parameters.

Q: What is validation?
A: Ensures data correctness.

Q: Explain session state.
A: Stores user data across requests.

Q: What is caching?
A: Stores data for faster access.

Web API & Security (15)

Q: What is Web API?
A: Framework for building HTTP services.

Q: Explain REST.
A: Architectural style using HTTP.

Q: What is JSON?
A: Lightweight data format.

Q: Explain JWT.
A: JSON Web Token for authentication.

Q: What is OAuth?
A: Authorization framework.

Q: Explain HTTPS.
A: Secure HTTP using SSL/TLS.

Q: What is CORS?
A: Cross-Origin Resource Sharing.

Q: Explain API versioning.
A: Managing multiple API versions.

Q: What is throttling?
A: Limiting API requests.

Q: Explain rate limiting.
A: Restricting request frequency.

Q: What is HMAC?
A: Hash-based message authentication.

Q: Explain CSRF.
A: Cross-Site Request Forgery attack.

Q: What is XSS?
A: Cross-Site Scripting attack.

Q: Explain basic auth.
A: Simple username/password authentication.

Q: What is bearer token?
A: Token-based authentication method.

Entity Framework & Database (15)

Q: What is EF Core?
A: ORM for .NET Core.

Q: Explain DbContext.
A: Primary class for EF operations.

Q: What is migration?
A: Updating database schema.

Q: Explain LINQ to Entities.
A: Querying database using LINQ.

Q: What is eager loading?
A: Load related data immediately.

Q: What is lazy loading?
A: Load related data on demand.

Q: Explain explicit loading.
A: Manually load related data.

Q: What is Code First?
A: Define model classes first.

Q: What is Database First?
A: Generate models from database.

Q: Explain concurrency.
A: Handling simultaneous data updates.

Q: What is transaction?
A: Atomic unit of work.

Q: Explain stored procedure.
A: Precompiled SQL statements.

Q: What is indexing?
A: Improves query performance.

Q: Explain normalization.
A: Organizing data to reduce redundancy.

Q: What is denormalization?
A: Combining tables for performance.

Front-End (10)

Q: What is HTML?
A: Markup language for web pages.

Q: Explain CSS.
A: Styles for HTML elements.

Q: What is JavaScript?
A: Language for client-side scripting.

Q: Explain DOM.
A: Document Object Model.

Q: What is AJAX?
A: Asynchronous JavaScript and XML.

Q: Explain responsive design.
A: Adapts layout to screen size.

Q: What is Blazor?
A: Framework for C# in browser.

Q: Explain SPA.
A: Single Page Application.

Q: What is TypeScript?
A: Superset of JavaScript.

Q: Explain WebAssembly.
A: Binary format for web apps.

Azure & Cloud Integration (10)

Q: What is Azure?
A: Microsoft cloud platform.

Q: Explain App Service.
A: Platform for hosting web apps.

Q: What is Azure Functions?
A: Serverless compute service.

Q: Explain Blob Storage.
A: Stores unstructured data.

Q: What is Azure SQL?
A: Managed SQL database.

Q: Explain Key Vault.
A: Securely store secrets.

Q: What is Azure DevOps?
A: Tools for CI/CD.

Q: Explain Service Bus.
A: Messaging between apps.

Q: What is CDN?
A: Content Delivery Network.

Q: Explain scaling.
A: Adjust resources based on demand.

System Design & Architecture (10)

Q: What is microservices?
A: Architecture of small services.

Q: Explain monolith.
A: Single-tiered application.

Q: What is load balancing?
A: Distribute traffic across servers.

Q: Explain caching strategy.
A: Improve performance using cache.

Q: What is CQRS?
A: Command Query Responsibility Segregation.

Q: Explain event sourcing.
A: Store state as events.

Q: What is API Gateway?
A: Entry point for APIs.

Q: Explain design patterns.
A: Reusable solutions to problems.

Q: What is SOLID?
A: Five principles of OOP design.

Q: Explain scalability.
A: Ability to handle growth.